

MATH 3113

Abstract Algebra I

Fall 2023

Professor John Bergdall
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MWF 9:40 - 10:30 AM
Classroom: SCEN 201

Overview

This course introduces abstract algebra. In algebra, we focus on mathematical structure. When are structures the same? Different? And how? We also focus on mathematical symmetry. How are objects internally built or organized?

Learning goals

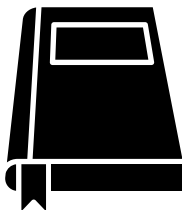
Recognize
algebraic
structures.

Understand
symmetry.

Analyze
mathematical
proofs.

Manipulate
algebraic
axioms.

Basic resources



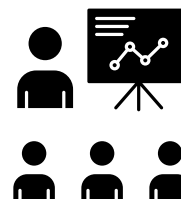
Textbook

Abstract Algebra (4th ed.)
by Beachy and Blair
(Rent/buy: \$47/\$71*)



Course materials

Materials posted to Blackboard.
Assignments on Gradescope.
(Free)



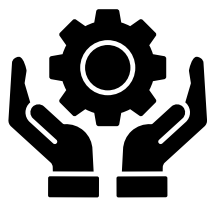
Office hours (SCEN 220)

Monday	10:35-11:30
Wednesday	2:00-3:00
Thursday	2:00-3:00

*Used costs. If costs cause hardship or would limit your course access, email me for support.

Tasks and grades

Course tasks are divided into *Supporting Tasks* and *Performance Tasks*. Supporting Tasks keep the course moving. Performance Tasks reflect your understanding. **Assignment and grade details on pages 3-5.**



Supporting Tasks

Reading Questions and Class Contributions.



Performance Tasks

Homework Questions and Mastery Assessments.

Turn-in

Homework Questions will be due Friday afternoons (5:00 PM). **Reading Questions** are due the start of the relevant class meeting class.

Revisions

Revising your work reinforces your learning. Each task may be revised *once*.

- Revisions for Reading Questions in a given week are due the **Friday of the following week**.
- Revisions for Homework Questions will be collected **on the in-class assessment dates**.

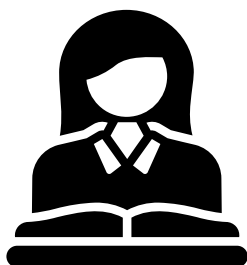
Late work

Completing tasks by their due dates keeps you on track. Nevertheless, you *may turn work in late*.

- Unanswered Reading Questions may be turned in through the revision process.
- Homework Questions may be turned in by **start of class on Monday**. Otherwise, they are revisions.
- It is *to your benefit* to turn in *anything* you have completed by the initial due dates!
- *Additional* deadline flexibility is possible. Contact me as early as possible to discuss a fair action.

Supporting Tasks

There are two main Tasks that support the course. **Reading Questions** check understandings before discussion. **Class Contributions** measure your impact on the learning environment.



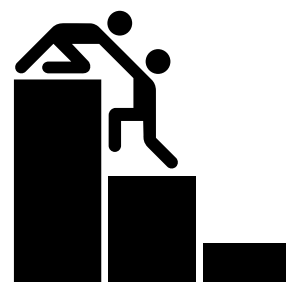
Reading Questions*

- **1-2 questions** due *per* reading. Correct answers = 1 point.
- **Initial response** timed and auto-graded on Gradescope.
- **Revisions** include additional, complete, reasoning.

*We will fill out the first Reading Questions *collectively* on Wed. August 23rd.

Class Contributions (a selection of options)

- **Discuss** on Blackboard.
- **Post** YouTube help.
- **Reinforce** a friend's ideas.
- **Ask** questions in class.
- **Answer** questions in class.
- **Correct** *my* materials.
- **Solve** further exercises.
- **Contribute** in office hours.
- **Participate** in course survey.
- See p. 7 for info on pts.



Supporting Task Grading

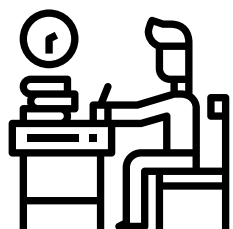
Reading Questions answers are worth one point. During revision, the point is earned for a complete explanation and half a point for *any* explanation. Class Contributions are always worth one point. Your Supporting Task score is gotten by dividing the earned points by the opportunity points.

Example

Sheri got 1 out of the first 5 Reading Questions correct. They then asked 2 questions on Blackboard. They also correctly revised 1 Reading Question. So, they've had 5 Question opportunities, 2 Contribution opportunities, and earned 4 points total. Their Supporting Tasks are worth $4/7 \sim 57\%$

<u>Sheri's Supporting Tasks</u>	Reading Questions	Class Contributions	Total
Initial	1 out of 5	0 out of 0	1 out of 5
Final	2 out of 5	2 out of 2	4 out of 7

Performance Tasks



Homework Questions substantially build upon specific topics. **Mastery Assessments** gauge your broader ideas on algebra and symmetry.

Homework Questions

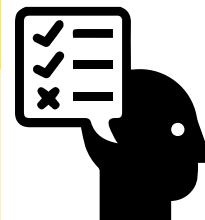
- **2-3 Questions** due *most* weeks on *Fridays*.
- **Collaborate!** As long as you attribute credit to your source(s).
- **Revisions** submitted with Mastery Assessments.

Dates:

Oct 2 - Oct 6
Oct 30 - Nov 3
Nov 27 - Dec 1
Dec 11 (Final)

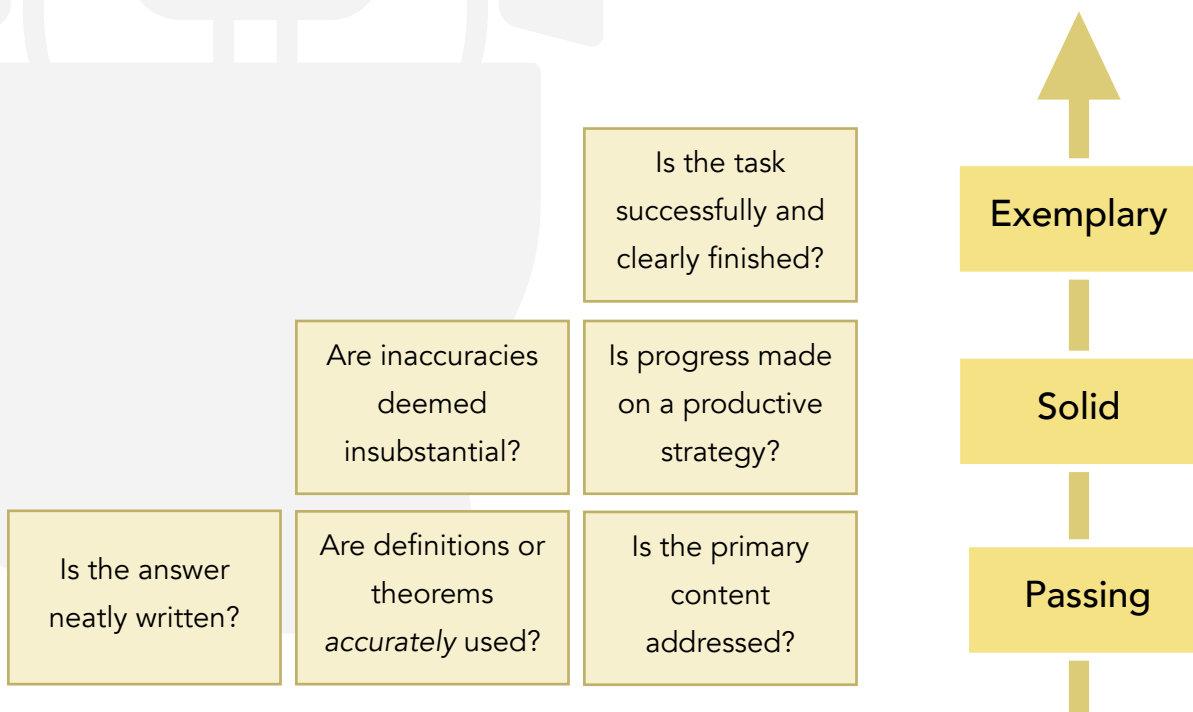
Mastery Assessments

- **Exam-like** with two components. **Out-of-class** tests *reading* skills. **In-class** tests *writing* skills. **Both** test *proving* skills.
- **Revisions** due within one week.



Performance Task Grading

Each Homework Question and Mastery Assessment Question is graded out of six points, based on qualities of *passing*, *solid*, and *exemplary* work. Note: your Homework and Assessments questions are *equally* weighted.



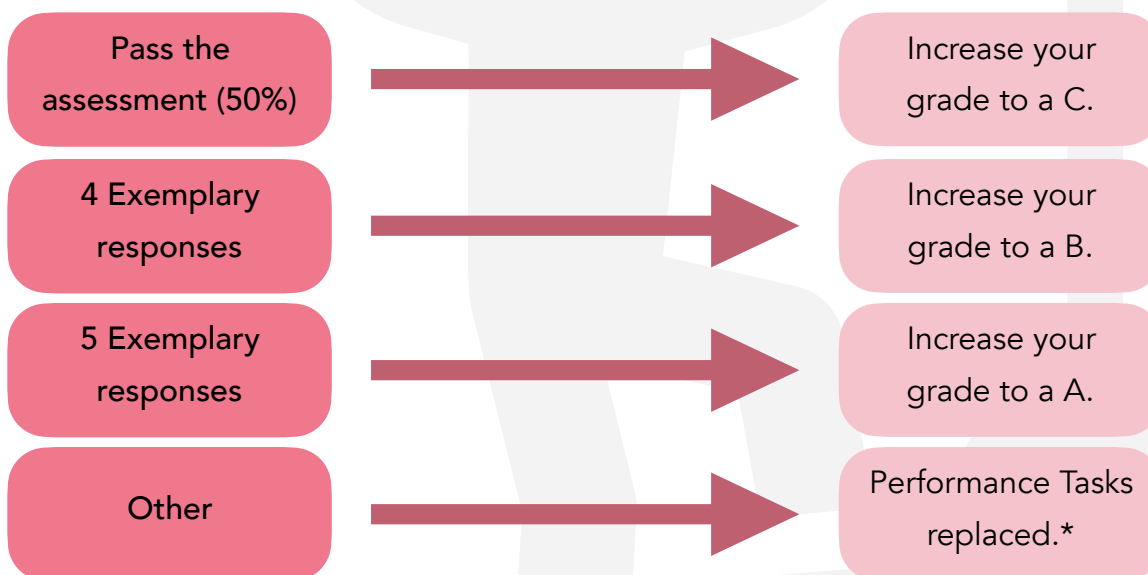
Final grade

Your final grade is determined by thresholds. We may make *more generous* thresholds later in the semester.

	To earn a C...	To earn a B...	To earn an A...
Supporting Tasks	at least 70%	at least 80%	at least 90%
Performance Tasks	at least 50%	at least 70%	at least 90%

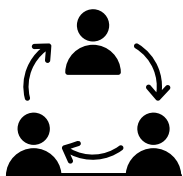
Final Mastery Assessment

The **optional final assessment** will consist of 6 questions over 2 hours. It is cumulative, but it can *only* raise your grade. Here is how!



*We replace low-scoring Performance Tasks with higher-scoring Final Mastery Assessment Questions.

Tips for success



Collaborate!
With peers.

Post! To
Blackboard.



Ask! Me early
and often.

Tips for the stuck and frustrated.

- Take a break. Walk away.
- Explain to someone why you are stuck.
- Check hypotheses or assumptions.
- Work out a single example.
- Keep going!

Further Resources

Appointments

Office hours will not always fit your schedule. Email me to set appointments. In your initial message, please explain (i) what you want to talk about, (ii) *your* availability and (iii) if meeting in person is important (Zoom is possible).



Email

Your questions are important. Email me or post to Blackboard. (I may ask you to do this anyways.) Responses may take 24 hours, but send a reminder if I do not reply.

Access

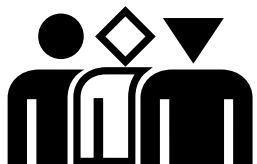
I want you to pass this class, with success. If you find resources not being provided, tell me and I will work with you to fix the issue. For some students, the Center for Educational Access (cea.uark.edu) can coordinate student accommodation requests. Please request your accommodation letter early semester and meet me to discuss it.



Class formation

Names and pronouns

You deserve to be called how you want. Let me know your preferred name and pronouns any time. When meeting one another, re-introduce yourself *frequently*. Everyone's names, pronouns, etc. should be known and properly used.



Diversity

I pledge attention to your identity and experience, regardless of age, background, beliefs, ethnicity, gender identity and expression, national origin, racial identity, religious beliefs, sexual orientation, and any other visible or non-visible categories. You are expected to do the same for myself and your peers.

Academic integrity

I trust my students to follow (i) the University of Arkansas Academic Integrity Policy and (ii) my own explicit assignment instructions. When you have questions about a boundary or rule, ask me rather than assuming you know the answer.



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Changes from prior versions (online only)

Changes 08-29-2023

- Included information on Reading Question revisions.
- Clarified Class Contribution list is only a list of *possibilities*. Explicitly included:
 - Visits in office hours. (When your visit leads to an impact on the class, such as me posting a discussion topic to Blackboard.)
 - Participation in course surveys.
- Regarding Late Work: added the point: *Additional deadline flexibility is possible. Contact me as early as possible to discuss a fair action.*

Changes 09-08-2023

- The exam dates were previously inaccurate. The first Mastery Assessment will take place Oct 2 - 6 and the second will take place Oct 30 - Nov 3. (The third assessment and final were accurate.)

Changes 10-04-2023

- For class contributions, a new weighting scheme is used. In-class contributions continue to count as +1 points. Out-of-class contributions are increased to +2 points, to encourage students often less rewarded by their contributions to the environment.

Changes 12-04-2023

- For the final assessment, we adjusted the impact of simply passing the assessment (you pass the course).